

ZILAIR, Russian Federation

WMO: 350260

Lat: 52.239N

Lon: 57.452E

Elev: 1709

StdP: 13.81

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.0	-14.0	-25.2	1.4	-19.4	-19.7	2.0	-13.1	16.1	15.2	14.0	14.2	2.2	30	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	86.4	63.2	83.2	62.7	80.1	61.5	66.1	80.2	64.7	78.7	63.2	76.5	5.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	86.0	71.0	59.5	80.9	69.5	58.0	76.6	68.6	31.7	80.1	30.6	78.8	29.5	76.7	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	15.7	13.3	11.5	DB	-27.0	91.1	7.1	3.1	-32.2	93.4	-36.3	95.2	-40.3	96.9	-45.5	99.1
(5)				WB	-27.0	68.6	7.1	2.5	-32.1	70.4	-36.3	71.8	-40.3	73.2	-45.4	75.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	7.9	10.5	21.9	39.4	53.4	61.2	64.7	62.5	51.1	38.0	22.9	11.4
	DBStd	22.59	10.06	10.62	8.84	10.18	8.06	8.14	6.86	7.86	8.08	8.73	10.65	11.33
	HDD50	6173	1306	1107	871	342	59	11	1	5	81	380	814	1196
	HDD65	10350	1771	1527	1336	768	366	164	91	142	423	837	1264	1661
	CDD50	1507	0	0	0	24	163	348	456	394	113	9	0	0
	CDD65	207	0	0	0	0	5	51	81	65	5	0	0	0
	CDH74	2536	0	0	0	4	133	680	878	769	72	0	0	0
	CDH80	676	0	0	0	0	10	196	225	232	13	0	0	0
Wind	WSAvg	4.5	4.0	4.9	5.4	5.3	4.9	4.5	4.3	3.9	3.7	4.4	4.6	4.2
Precipitation	PrecAvg	20.4	1.5	1.4	1.8	1.7	1.6	1.9	1.9	1.6	1.4	1.9	1.9	1.8
	PrecMax	38.5	3.8	3.9	6.0	4.3	3.7	4.6	4.1	6.7	3.9	4.2	4.2	4.0
	PrecMin	13.1	0.0	0.1	0.0	0.2	0.2	0.0	0.3	0.2	0.1	0.0	0.1	0.1
	PrecStd	5.2	0.9	0.9	1.3	1.1	1.0	1.1	1.2	1.3	1.1	1.1	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.9	33.8	48.3	72.8	81.2	89.3	90.2	90.2	81.2	66.5	46.2	33.0
		MCWB	30.2	30.1	38.4	52.7	58.6	63.4	64.3	64.0	57.4	50.8	43.3	31.4
	2%	DB	27.6	30.3	41.7	67.6	77.6	85.3	85.4	86.1	74.8	61.3	41.8	30.7
		MCWB	26.6	28.3	34.9	50.7	57.2	63.0	64.0	62.7	55.9	47.9	39.3	29.5
	5%	DB	25.7	28.0	37.0	61.4	74.5	81.7	82.2	82.4	70.3	56.5	37.8	29.0
		MCWB	24.9	26.4	32.3	47.7	56.5	62.1	63.4	62.0	54.6	45.7	35.9	27.8
	10%	DB	22.7	25.5	33.9	55.7	70.3	77.4	79.0	78.3	65.7	52.0	35.0	26.7
		MCWB	21.8	23.9	30.8	44.9	54.4	60.5	62.3	61.3	52.9	44.3	33.3	25.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.3	31.5	40.3	54.4	62.0	66.6	68.5	67.2	60.6	52.8	44.2	31.9
		MCDB	30.5	32.5	47.2	69.5	74.9	81.8	80.4	81.5	74.7	62.0	45.1	32.4
	2%	WB	27.0	28.8	35.5	51.3	59.5	64.8	66.7	64.9	57.9	49.8	39.6	29.7
		MCDB	27.6	29.9	40.4	64.5	73.2	80.6	79.9	79.9	70.0	58.2	41.2	30.4
	5%	WB	24.9	26.6	33.2	48.7	57.7	63.3	65.4	63.6	55.9	47.1	36.3	28.0
		MCDB	25.6	27.9	36.3	59.9	71.3	78.2	78.4	78.3	66.5	54.5	37.7	28.9
	10%	WB	21.9	23.9	31.2	45.8	55.8	61.7	63.8	62.2	54.0	44.8	33.7	25.6
		MCDB	22.7	25.5	33.8	54.7	67.8	74.8	75.8	75.7	63.9	50.9	34.7	26.5
Mean Daily Temperature Range	5% DB	MDBR	13.5	16.5	16.2	17.9	22.2	22.1	21.0	21.9	20.1	15.1	10.3	12.9
		MCDBR	10.0	16.9	19.3	26.9	28.8	28.1	25.8	27.6	28.6	25.9	12.5	10.8
		MCWBR	9.7	14.2	13.5	13.6	12.2	10.7	9.7	9.4	13.1	15.0	10.1	10.0
	5% WB	MCDBR	9.6	13.0	15.1	24.2	25.6	24.1	21.7	24.2	24.0	22.5	10.7	9.9
		MCWBR	9.5	11.5	11.1	13.2	12.0	10.1	9.4	9.7	12.3	14.3	9.4	9.5
Clear-Sky Solar Irradiance	taub	0.282	0.309	0.364	0.412	0.394	0.387	0.399	0.419	0.368	0.342	0.311	0.279	
	taud	2.089	2.048	1.956	2.054	2.255	2.338	2.310	2.231	2.341	2.314	2.185	2.076	
	Ebn at Noon	217	247	254	259	272	275	269	255	256	237	205	193	
	Edh at Noon	24	34	46	47	41	38	38	39	31	26	22	20	
All-Sky Solar Radiation	RadAvg	351	702	1130	1432	1784	1989	1905	1579	1083	557	253	229	
	RadStd	49	68	126	139	159	186	133	133	130	107	49	45	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+2.29	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page